

Playwright Capstone Project

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TECHNOLOGY STACK

Technology	Purpose	Usage
Playwright + JavaScript	Automation	Search, filters, cart workflows
Node.js	Runtime	Dependency management
GitHub Actions	CI/CD	Workflow execution
HTML Reports	Reporting	Debugging and evidence
Cross-browser	Compatibility	Chromium/Firefox validation

FUNCTIONAL AREAS COVERED

- Search workflows validate keyword input and suggestion behavior.
- Filter scenarios verify category, price, and brand refinement accuracy.
- Cart operations examine quantity changes and persistence after refresh.
- Coupon validation checks promotional availability and behavior.
- API monitoring tracks network responses linked to search interactions.

FRAMEWORK CAPABILITIES

- Reusable Page Object Model simplifies maintenance for dashboard journeys.
- Dynamic locators support changing product layouts and filters.
- Parallel execution reduces total regression time.
- Screenshots capture failures during cart or search validations.
- Trace generation provides detailed execution evidence.

PRIMARY OBJECTIVES

- Reduce repetitive manual testing for common customer journeys.
- Increase regression speed for search, filter, and session scenarios.
- Improve defect detection through assertions and API interception.
- Build reusable automation assets supporting long-term automation maintenance.

- Strengthen reporting visibility for enterprise-scale testing.

AUTOMATION DASHBOARD

Journey	Scenario	Playwright Skill
Search	Search by keyword	Assertions + locators
Filters	Brand/price/category	Dynamic locators
Product	Open PDP & validate	POM
Wishlist	Save item	State checks
Cart	Add/remove/update qty	Assertions
Coupons	Promo availability	Validation
Session	Refresh retains cart	Storage
API	Monitor search responses	Intercept
Responsive	Chromium/Firefox	Parallel
Performance	Page load timing	Metrics

EXPECTED OUTCOMES

The framework is expected to accelerate regression cycles, improve debugging quality, and increase confidence in repeated execution across supported browsers.

PROJECT CONCLUSION

This capstone demonstrates practical enterprise automation through maintainable architecture, realistic workflows, and execution reporting.